

Arthur Wang

✉ yankesswang@gmail.com |  arthur-wang0413 |  arthurwang0413 | 📍 Taiwan

SUMMARY

Applied AI Engineer specializing in **on-device AI agents, RAG systems, local LLM applications, and multimodal AI products**. Experienced in turning LLM prototypes into production-ready desktop and enterprise AI systems, covering retrieval pipelines, knowledge graphs, agent workflows, evaluation harnesses, and local model deployment. Strong end-to-end ownership across product planning, system architecture, engineering implementation, model evaluation, and cross-functional delivery.

WORK EXPERIENCE

GIGABYTE Technology

Jun. 2024 – Present

AI Algorithm Deputy Manager / Technical Lead

New Taipei, Taiwan

- Led technical development of GIGABYTE's on-device AI and enterprise AI systems, delivering **6+ AI applications** across local AI agents, hardware Q&A, laptop assistants, translation, and meeting intelligence.
- Designed local LLM deployment architectures using **llama.cpp, vLLM, LangChain, and LlamaIndex**, enabling offline-capable AI features under approximately **4GB VRAM** constraints.
- Built enterprise RAG and knowledge graph systems over **1,000+ product specification documents**, reducing FAE lookup time from **5–10 minutes to under 30 seconds** in internal workflows.
- Developed AI Agent workflows covering task planning, memory, tool invocation, scheduling, permission control, and execution tracking to improve reliability in real product environments.
- Established evaluation pipelines for retrieval quality, answer correctness, hallucination analysis, and model performance tracking, achieving **92–97% accuracy** on internal benchmark datasets.
- Integrated OCR, ASR, translation models, VLM pipelines, and LLM summarization to support document translation, real-time transcription, and meeting intelligence across **55 languages**.
- Collaborated with product, engineering, FAE, customer service, and management teams to ship AI capabilities into laptop-side applications, internal support systems, and hardware product workflows.

Tarobo Investment

Jun. 2023 – Oct. 2023

Quantitative Trading Engineer

Taipei, Taiwan

- Developed deep learning models for stock price direction prediction, improving directional prediction accuracy by **15%**.
- Built a Python-based backtesting pipeline for historical tick data, reducing strategy evaluation time from **6 hours to 15 minutes**.
- Conducted systematic backtesting and signal analysis to support quantitative trading strategy development.
- Analyzed market signals, financial indicators, and model outputs to evaluate trading opportunities and improve strategy iteration.

KEY PROJECTS

Gimate Agent: On-Device AI Agent Built on OpenClaw

Feb. 2026 – Present

- Built an on-device laptop AI Agent integrating **OpenClaw, Windows MCP, local small language models, and agent skills** for system control, personalized assistance, and workflow automation.
- Designed the Agent Orchestration Core, using **planning-with-files and harness engineering** to cover task planning, memory, tool invocation, permission control, scheduling, logging, and execution tracking.
- Developed Hardware Intelligence capabilities for monitoring CPU, RAM, processes, battery, and network status, enabling performance diagnosis and system optimization suggestions.
- Integrated GiMate API and long-term memory to learn user preferences and personalize fan speed, performance modes, power consumption, and thermal behavior.
- Extended the agent with personal knowledge base and assistant capabilities, including file search, cross-document summarization, calendar, Gmail, reminders, and proactive task support.

MemKG-Retrieve: Evidence-Grounded Long-Term Memory for AI Agents

Oct. 2025 – Present

- Designed a long-term memory framework for Personal Agents combining **knowledge graphs, hybrid retrieval, reranking, and evidence-grounded answer generation**.
- Built a retrieval evaluation harness covering BM25, vector retrieval, RRF reranking, and evidence selection strategies as reproducible experiments.
- Designed a hybrid memory retrieval pipeline improving recall of time-series events, entity relationships, and cross-passage memories; applied agent grep / evidence extraction to surface key summaries and supporting evidence for LLM responses.
- Built an Auto Research Agent to generate hypotheses, design experiments, compare retrieval architectures, and identify optimal memory system configurations based on benchmark results.
- Achieved **82% overall accuracy** on the LoCoMo benchmark, outperforming Mem0's reported **68.4%**, demonstrating improved long-term memory retrieval and evidence-grounded response quality.

FAE Support System: Hybrid RAG for Hardware Q&A

Jan. 2026 – Present

- Built a Hybrid RAG hardware Q&A system covering GIGABYTE product lines, supporting specification lookup, compatibility checks, and troubleshooting for motherboards, GPUs, and laptops.
- Established a spec-to-structured-data pipeline parsing and standardizing **1,000+ product documents** into a Neo4j hardware knowledge graph.
- Deployed BERT-based query classification and entity linking to identify product models, categories, and ambiguous hardware references from user questions.
- Combined graph-based retrieval, BM25, embeddings, and fuzzy matching to support exact model queries, fuzzy product descriptions, and natural language troubleshooting.
- Achieved approximately **95% model identification accuracy, 97% Top-3 entity linking hit rate, and 92% hardware Q&A accuracy** on internal test sets.
- Reduced FAE specification and compatibility lookup time from **5–10 minutes to under 30 seconds**, cutting manual search effort by approximately **70%**.

NB Tutor: On-Device AI Laptop Assistant

Jul. 2025 – Present

- Built a local AI support assistant for GIGABYTE laptops using Qwen-series small language models deployed via **llama.cpp**.
- Integrated product specification retrieval, GiMate feature Q&A, bilingual English/Chinese responses, and technical support guardrails.
- Designed brand-aligned response guidelines to ensure answers matched support policies, product communication standards, and safety requirements.
- Built an automated evaluation pipeline using laptop specifications and utility feature documents to validate answer quality across **20 laptop models**.
- Developed error analysis workflows to track retrieval misses, hallucinations, and model response failures for continuous RAG optimization.

Translate Engine: Multimodal Translation System

Jan. 2026 – Present

- Built a local translation system using **TranslateGemma-4B**, supporting **55 core languages** with approximately **3–4GB VRAM** deployment requirements.
- Integrated a VLM pipeline using **Qwen3-VL-2B-Instruct** to process images, scanned PDFs, and formatted documents while preserving tables, paragraphs, and layout structure.
- Improved small-model translation stability through chunking strategy, max output token control, and prompt engineering to reduce language confusion, truncation, and repetition loops.
- Designed a sequential GPU loading mechanism to stage-load VLM/OCR and translation models, preventing VRAM stacking and keeping each stage within approximately **4GB VRAM**.
- Built the desktop application and standalone executable using **Electron, FastAPI, and PyInstaller**, lowering the barrier for local deployment.

Meeting Intelligence System: Real-Time Transcription, Translation, and Summarization

Jan. 2026 – Present

- Built a meeting intelligence pipeline that converts raw meeting audio into structured summaries, including key points, decisions, action items, and follow-ups.
- Integrated ASR, translation, and LLM summarization modules to support cross-language meeting transcription and summarization.

- Evaluated and integrated **Whisper** and **Qwen ASR**, achieving Chinese CER of **5%** and English WER of **2%** in internal mixed-language meeting tests.
- Integrated VAD to detect speech boundaries, automatically segment audio, and improve real-time transcription smoothness.
- Designed a streaming transcription pipeline supporting live transcript updates during recording.

EDUCATION

National Chiao Tung University
Bachelor of Science in Computer Science

Sep. 2017 – Jun. 2021
Hsinchu, Taiwan

PROFESSIONAL SKILLS

AI Agents & LLM Applications : Agent Workflows, Tool Use, Memory Systems, RAG, Evaluation Harnesses, Prompt Engineering

Retrieval & Knowledge Systems : Hybrid Search, Knowledge Graphs, BM25, Vector Search, Reranking, Entity Linking, Neo4j, FAISS, Elasticsearch

Local LLM & Model Optimization : llama.cpp, vLLM, Quantization, QLoRA, GRPO, On-device Inference

Multimodal AI : OCR, ASR, VLM Pipelines, Translation, Meeting Summarization

Frameworks : LangChain, LlamaIndex, PyTorch, HuggingFace Transformers, OpenAI API

Application Development : FastAPI, Electron, REST API, PyInstaller, Desktop App Packaging

Languages : Mandarin Chinese (Native), English (Intermediate)